# Automation Basics with n8n

Instructor guide · Day 3 · Session 5

Open the trial together; triggers, actions and items; the leave request as a live workflow; build the request-intake lab; test, then publish

## Session at a glance

| | |
|---|---|
| **Day / session** | Day 3 · Session 5 · 120 minutes |
| **Lead instructors** | Mousa AL-Akhras, Mohammed Abdelmajeed |
| **Online module** | https://albanna-tutorials.com/ai-administrator/05_automation_basics_n8n.html |
| **Deck** | Day3_S5_Automation_Basics_with_n8n.pptx, 50 slides, speaker notes included |

## Learning objectives

- Open an n8n Cloud trial and know its limits: 14 days, 1,000 executions, export your workflows every day.
- Describe what an automation platform does and compare n8n, Make, Zapier and Power Automate.
- Tell a trigger from an action and explain how items and fields flow from node to node.
- Read a workflow against the Session 4 future-state map: which node implements which step.
- Build a Form Trigger, Google Sheets, IF and Gmail workflow, test it step by step and read each node's output.
- Connect credentials safely, add Retry on Fail and an error workflow, and publish only after testing.

## Run of show

| Minutes | Block | Focus |
|---|---|---|
| 5 to 20 | 1. Open n8n together | Same clicks, same minute, everyone |
| 20 to 35 | 2. What an automation platform is | A trigger, then actions, with data flowing between them |
| 35 to 55 | 3. Demo: the leave request runs | Session 4's first build, as a real workflow |
| 55 to 60 | Break | Five minutes. Keep your n8n tab open. |
| 60 to 100 | 4. Lab: request intake | Build it yourself, in breakout rooms, 40 minutes |

| | | |
|---|---|---|
| 100 to 108 | 5. Make it safe, then publish | Retry, an error workflow, and the difference between test and production |
| 108 to 118 | Kahoot | 20 questions on today |

The welcome, objectives and run-of-show slides take the first minutes before Section 1. The break is planned at the midpoint (see the run-of-show slide in the deck). Cuts if running behind are listed under Troubleshooting.

## Materials checklist

- Online session on Zoom. The presenting instructor is host; the second instructor is co-host (Participants, More, Make co-host) and runs the chat, the breakout rooms and the 'stuck' room during sign-up and the lab.
- Zoom settings, once, in the Zoom web portal (Settings, Meeting): Breakout room on; Chat on; Non-verbal feedback (Raise hand) on; 'Allow participants to share in breakout rooms' on, so the co-host can see a participant's n8n screen.
- Share Screen of this deck (Day3_S5_Automation_Basics_with_n8n.pptx) in PowerPoint slide show with Presenter View on a second screen; for the demo share a separate browser window with the n8n editor, zoomed to 125%.
- Instructor n8n Cloud workspace prepared the day before (Day3_S5_Demo_Setup_Guide.pdf): the leave demo imported, Google Sheets and Gmail connected, three test runs done, test rows cleared.
- Google Sheet made from Day3_S5_Leave_Demo_Sheet.xlsx (File, Save as Google Sheets) open in a second tab, on the Requests tab.
- Files for the chat: Day3_S5_n8n_Signup_Sheet.pdf (minute 5), Day3_S5_Lab_Handout.pdf and Day3_S5_Requests_Template.xlsx (minute 60), Day3_S5_Request_Intake_Solution.json (after the session only).
- Kahoot imported from Day3_S5_Kahoot.xlsx and ready to launch (PIN on screen at minute 108); answer key Day3_S5_Kahoot_Answer_Key.pdf. The written Quiz bank (Day3_S5_Quiz_Bank.pdf) stays available for homework or revision.
- No bot this session: @aicaravan_bot is not used and must not be switched.
- Three narrated videos in 06_Videos (real n8n screens, subtitles in .srt): Video1 Open Your Workspace (1:26) for late joiners or anyone stuck at sign-up; Video2 Leave Request Demo (2:09) as the fallback if the live demo fails; Video3 Lab Request Intake (2:25) to share with the lab handout. Play with Zoom Share sound and Optimize for video clip.
- Participants: an e-mail address they can open during the session (for the n8n verification mail), a Google account with Sheets and Gmail (the same account for both), a laptop (n8n does not work well on a phone), and their M4-future-state file from Session 4.

## Before the session

- Read the live module once (link above); the lab slides follow its eight lab steps, with two corrections found while testing (see the two 'trap' slides).
- The day before: follow Day3_S5_Demo_Setup_Guide.pdf on your own n8n trial. Open your trial no more than a few days before the session, so it is still active on the last n8n day.
- Run the three demo tests once (pass, short notice, over balance) and check the Requests tab and your inbox; then delete the test rows so the sheet is empty for the live demo.

- Import Day3_S5_Kahoot.xlsx into Kahoot (Create, Import spreadsheet) and do a one-question test game.
- Refresh the 'Latest reading' slide if something newer has appeared.

## Live sign-up and demo (minutes 5 to 20 and 35 to 55)

1. Minute 5: the co-host posts n8n.io/pricing and the sign-up sheet in the chat. Do the sign-up yourself on screen at the same pace (a fresh browser profile works), so participants copy each click.
2. Minute 12: the co-host opens breakout room 'Stuck at sign-up' and moves anyone with a missing verification e-mail or a blocked workspace there.
3. Minute 35: in your prepared workspace open 'S5 Demo - Leave request'. Point at each node and say which Session 4 step it is (slide 'One step, one node').
4. Minute 40: double-click the form node, show Test URL versus Production URL, click Execute workflow, open the test form in a new tab and submit Omar Khalil, 15 to 17 November (passes).
5. Minute 44: back in the editor, open each node and read its output panel aloud: 1 item in, 1 item out. Show the decision card arrive in your inbox.
6. Minute 48: run twice more: Sara Nasser, one day the day after tomorrow (short notice) and Lina Haddad, 15 to 17 November (only 2 days left). Show the false branch, the Reason column in the sheet and the Executions tab.
7. Minute 53: open Settings, Retry on Fail on a Gmail node, and say what the session lab adds: guard, retry, error workflow.

## Slide-by-slide talking points

Slide numbers match the deck. Quiz questions occupy two slides each (question, then answer).

**1. Automation Basics with n8n** · **Title**
Welcome to Day 3. One-sentence promise: by the end of these two hours everyone has an n8n workspace, has watched the leave request from Session 4 run as a real workflow, and has built and tested their own first automation. Still no code for participants.

**2. Where this session sits** · **Bullets**
Ask everyone to have two things ready: an e-mail inbox they can open now, and their M4-future-state file. We open n8n together in five minutes.

**3. What you will be able to do after this session** · **Bullets**
The deliverable is the lab: a published request-intake workflow with three test runs, plus the exported JSON.

**4. Run of show (120 minutes)** · **Table**
Ten seconds on screen. The lab is in the session this time, not homework: forty minutes in breakout rooms.

**5. 1. Open n8n together** · **Section** · **minute 5**
The co-host posts n8n.io/pricing and Day3_S5_n8n_Signup_Sheet.pdf in the chat now. You sign up on screen at the same pace.

**6. Sign up in six clicks** · **Steps**
Go slowly; do every click yourself on screen. At minute 12 ask for a Raise Hand from anyone who is not yet on the 'Let's build your first automation' page; the co-host takes them to the breakout room.

**7. What you see while the workspace starts** · **Figure · figure: d3s5_shot_01_instance_ready**
The animated preview below the button is a sample workflow; ignore it for now. Some participants will see the AI Assistant ask questions first; skipping is fine.

**8. The trial clock is always on screen** · **Figure · figure: d3s5_shot_02_trial_onboarding**
Point at the counter. 1,000 executions is plenty for the course if nobody publishes a schedule that runs every minute.

**9. Three rules for the trial** · **Callout**
Make this the one thing everyone remembers. We repeat it at the close of every n8n session. If the course ran past 14 days, exported JSON is what lets them continue in a new trial.

**10. Your daily habit: Export JSON** · **Figure · figure: d3s5_shot_07_menu_export**
Show where the menu is now, even with an empty workflow. Settings in the same menu is where the error workflow is chosen later.

**11. 2. What an automation platform is** · **Section · minute 20**
Fifteen minutes of concepts, each tied to a screen they now have open.

**12. Four platforms, one idea** · **Table**
The module has the full table. The skills transfer: every one of these has triggers, actions, data mapping, conditions and credentials. We use n8n because it can also run on your own server, which matters for data that must not leave the organisation.

**13. Where you are now** · **Figure · figure: d3s5_shot_03_overview**
Everyone clicks Build a workflow now. We do not use the AI builder today: you should know what each node does before you let anything generate one.

**14. Every workflow starts with one trigger** · **Figure · figure: d3s5_shot_05_trigger_picker**
Read the list. The trigger answers 'when does this run?'. Today's trigger is On form submission.

**15. Trigger and actions** · **Two columns**
A second event needs a second workflow, or a trigger that covers both. Quiz question.

**16. Data moves as items** · **Figure · figure: d3s5_items_flow**
The single most useful idea in n8n. Many 'it sent 40 e-mails' accidents come from forgetting it.

**17. An item is a set of fields** · **Figure · figure: d3s5_shot_13_form_output_json**
This is the real output of the leave form. Point at '1 item' above the JSON. Fields are what you drag into the next node.

**18. Credentials: sign in once, reuse everywhere** · **Figure · figure: d3s5_shot_15_gmail_credential**
Three rules from the module: least privilege, a shared or service account for production work, never paste a key into a node field or a sticky note.

**19. The trap we hit while preparing: 403 Forbidden** · **Figure · figure: d3s5_shot_19_error_403_no_permission**
Real error from preparing this session. Two causes: a different Google account, or an unticked box on Google's consent screen. Fix: open the credential, Switch account, same account as the sheet, tick every box. And the sheet must be a native Google Sheet, not an uploaded .xlsx.

**20. 3. Demo: the leave request runs** · **Section · minute 35**
Switch to the browser window with the prepared workspace. Participants watch; they do not build this one.

**21. One step, one node** · Figure · figure: d3s5_map_to_nodes
This is the bridge from Session 4. Ask the chat: which Session 4 candidate is this? (Candidates 2 and 3: look up the balance, run the policy checks, prepare the decision card.)

**22. The workflow, before credentials** · Figure · figure: d3s5_shot_06_leave_canvas_unconnected
Imported from Day3_S5_Leave_Request_Demo.json. The sticky note says what the workflow is for; a good habit.

**23. The trigger: an n8n form** · Figure · figure: d3s5_shot_09_form_node
Click Production URL to show the second address. The form is hosted by n8n; nothing to install.

**24. What the employee sees** · Figure · figure: d3s5_shot_10_test_form
Submit Omar Khalil, Annual leave, 15 to 17 November, your own address as manager. Then go back to the editor.

**25. The policy checks (written once by the instructor)** · Figure · figure: d3s5_shot_17_code_policy_checks
Keep this short. The point is that every rule here is one sentence from the leave policy (HR-POL-02): exactly the green steps of Session 4. Session 6 adds the amber step (reading the free-text note).

**26. The condition: All checks pass?** · Figure · figure: d3s5_shot_14_if_node_empty
Point at 'Convert types where required'. Then run the workflow.

**27. Run 1: checks pass** · Figure · figure: d3s5_shot_20_run_pass_green
Open your inbox and show the decision card. The manager still decides: the workflow prepared, it did not approve.

**28. Runs 2 and 3: flagged for HR** · Figure · figure: d3s5_shot_21_run_flagged_executions
Sara asks for tomorrow with one day's notice; Lina asks for 3 days with 2 left. The Executions tab is the audit log: click any run to see exactly what happened.

**29. The record** · Figure · figure: d3s5_shot_23_sheet_rows_status
The sheet stands in for the HR portal. Note what changed from Session 3: no re-typing, the checks are done before the manager sees anything, and HR only sees the exceptions.

**30. What the demo deliberately does not do** · Callout
Tie back to the prioritisation matrix: we built the high value, low risk candidates first.

**31. Break** · Section · minute 55
The co-host prepares breakout rooms of three or four, mixing faster and slower participants.

**32. 4. Lab: request intake** · Section · minute 60
The co-host posts Day3_S5_Lab_Handout.pdf and Day3_S5_Requests_Template.xlsx. Open the rooms at minute 63.

**33. Lab steps 1 to 4** · Steps
Steps follow the module. Rooms of three or four: one shares their screen, all build in their own workspace.

**34. Lab steps 5 to 8** · Steps
Step 6 is corrected from the module text; next slide. 'Activate' in the module is 'Publish' in today's n8n.

**35. Trap 1: an empty number is 0, not empty** · Callout
We found this by running the lab ourselves. It is a good lesson in itself: test the empty case, do not assume. The module text will be updated to match.

**36. The form participants build** · Figure · figure: d3s5_shot_24_lab_form
If a participant makes every field required, the guard never fires. That is fine in real life; for the lab, leave Amount optional.

**37. The finished lab, run 3: incomplete** · Figure · figure: d3s5_shot_26_lab_run_incomplete
The solution file Day3_S5_Request_Intake_Solution.json is shared after the session, not during the lab.

**38. Three tests, three rows** · Figure · figure: d3s5_shot_25_lab_sheet_rows
This is what each room should show the co-host before minute 100.

**39. In your breakout room** · Bullets
The two instructors rotate through the rooms. Most common issues: wrong Google account (403), uploaded .xlsx, forgetting to click Execute before submitting the test form, IF set to String instead of Number.

**40. 5. Make it safe, then publish** · Section · minute 100
Back in the main room. Eight minutes.

**41. Retry on Fail** · Figure · figure: d3s5_shot_16_retry_on_fail
'On Error' below it decides what happens after the last try: stop the workflow (default) or continue.

**42. When it still fails: the error workflow** · Two columns
An automation that fails silently is worse than no automation: people stop checking because they think it works.

**43. Test, then publish** · Two columns
The module and older tutorials say 'Activate'; today's n8n says Publish. Same idea.

**44. Kahoot** · Section · minute 108
Launch the kahoot imported from Day3_S5_Kahoot.xlsx; PIN on screen.

**45. Join the Kahoot** · Two columns
Answer key with all 20 questions: Day3_S5_Kahoot_Answer_Key.pdf.

**46. Deliverable** · Callout
Anyone who did not finish completes it before Session 6; the solution JSON is shared after the session.

**47. Before you close the tab** · Callout
Wait twenty seconds while they do it. Ask for a thumbs-up reaction in Zoom when done.

**48. Key takeaways** · Bullets
Preview Session 6: the same request intake gets an AI step that classifies the description, with an 'unsure' route to a person.

**49. Latest reading: n8n and automation at scale (2026)** · Latest articles
Optional two minutes or self-study. Checked on 11 October 2026; instructors may swap in newer pieces, keeping source, date and a one-line takeaway tied to the session.

**50. References** · References
Leave on screen during questions. All n8n screens are real screenshots; AI Caravan Ltd., its employees and its leave policy are fictional.

## Demo setup summary (instructor n8n workspace)

- Instructor's own n8n Cloud trial, opened a few days before the session. Import workflows/Day3_S5_Leave_Request_Demo.json (menu, Import, From file).
- A native Google Sheet with tabs Balances and Requests (from Day3_S5_Leave_Demo_Sheet.xlsx, File, Save as Google Sheets), owned by the same Google account you connect in n8n.
- Connect Google Sheets and Gmail with 'Sign in with Google', ticking every permission box. Paste the sheet link into the three Sheets nodes; change 'Flag to HR' to your own address.

- Three test runs (pass, short notice, over balance), then delete the test rows. No bot, no server, no API keys in this session.

Full step-by-step instructions, persona text and fallbacks: **Day3_S5_Demo_Setup_Guide.pdf** (in this package).

## Troubleshooting and cuts

| If | Then |
|---|---|
| The n8n verification e-mail does not arrive | Check spam and the Promotions tab; wait two minutes; then sign up with another address. The co-host handles this in the 'Stuck at sign-up' room so the main room keeps going. |
| A participant has used the n8n trial before with the same e-mail | Use another e-mail address (a personal Gmail is fine). One trial per address. |
| 'Sign in with Google' does nothing | The browser blocked the pop-up. Allow pop-ups for app.n8n.cloud (address bar icon) and click again. |
| Google Sheets node: 'Could not load list, 403 Forbidden, the caller does not have permission' or an empty document list | The Google account connected in n8n is not the account that owns the sheet, or a box was left unticked on Google's consent screen. Open the credential, Switch account, choose the sheet owner's account and tick every box. Seen live while preparing this session. |
| The document is an uploaded .xlsx (green 'XLSX' badge next to its name) | n8n's Sheets node reads native Google Sheets only. In Sheets: File, Save as Google Sheets, then use the new link. |
| The form says 'Form Submitted' but nothing appears in the editor | The test URL only listens after you click Execute workflow (or Execute step on the trigger), and only for one submission. Click it again before each test. |
| IF compares wrongly, for example 900 is 'smaller' than 500 | The value is text, not a number. Choose the Number operator (is greater than), or turn on 'Convert types where required' in the IF node. |
| An empty Amount passes the 'not empty' guard and is logged as 0 | Expected: n8n forms send 0 for an empty number field (verified 11 October 2026). Use 'Amount is greater than 0' as the guard (lab step 6, corrected). |
| Publish is greyed out or refuses | From n8n 2.43 (released 6 October 2026; the course workspace still ran 2.42.6 on 11 October) a workflow with an unconnected required input cannot be published. On any version, open the node with the red triangle and fill the missing field or credential. |
| A participant's connection drops | Their workflow is saved automatically in n8n; they rejoin, reopen the workflow and continue. Post the lab handout again in the chat. |
| Running behind at the break | Skip the second and third demo runs (show the Executions tab instead) and shorten the platform table to one sentence. Do not cut the sign-up, the lab or the export reminder. |

## Latest articles used on the slides

- **n8n 2.43.0 release**. n8n release notes, 6 October 2026. docs.n8n.io/release-notes
  Stronger guardrails and better publishing: a workflow with unconnected required inputs can no longer be published. The red triangles you saw today now block Publish.
- **No-Code Workflow Automation at Scale for IT Operations Teams**. Intellistack, 16 September 2026. intellistack.com/resources/blog/no-code-workflow-automation-at-scale-for-it-operations-teams
  When building outruns governance you get workflow sprawl and unknown owners. Why every automation needs an owner (Session 4) and an error workflow (today).
- **In The Loop 2026, n8n's first conference**. n8n, 13 to 14 October 2026, Berlin. n8n.io/intheloop
  Talks and workshops on production automation and agents, this week. Recordings are a good follow-up to Sessions 5 and 6.

## References


1. Al-Banna, A. (2026). AI Administrator, Module 5: Automation Basics with n8n. albanna-tutorials.com/ai-administrator/05_automation_basics_n8n.html
2. n8n documentation: n8n Form Trigger node. docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.formtrigger
3. n8n documentation: Google credentials and the Google Sheets node. docs.n8n.io/integrations/builtin/credentials/google
4. n8n documentation: Error handling, Error Trigger and Retry on Fail. docs.n8n.io/flow-logic/error-handling
5. n8n documentation: Data structure, items. docs.n8n.io/data/data-structure
6. n8n release notes 2.43.0 (6 October 2026): publishing blocked for workflows with unconnected required inputs. docs.n8n.io/release-notes
7. n8n Cloud pricing and free trial (14 days, 1,000 executions). n8n.io/pricing
8. Al-Banna, A. (2026). AI Administrator, Module 4: Workflow Mapping II, Designing the Future State (the leave-request future state used in the demo).
9. IEEE CS Region 8 AI Caravan (2026). AI Administrator Track Curriculum. aicaravan.org


Screenshot: n8n Cloud (n8n GmbH), captured 11 October 2026 on the course trial workspace. Every n8n screen in this deck is a real screenshot taken on 11 October 2026 on the instructors' n8n Cloud trial (version 2.4x, the 'Publish' editor). Two conceptual figures were drawn for the course (tools/make_day3s5_figures.py). The running example is the fictional AI Caravan Ltd. leave process from Sessions 3 and 4; all names in the screenshots are fictional.